



## WEEKLY LESSON PLAN

# Renewable Energy & Simple Circuits

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Grade 6 — Science / STEM



Week of October 12–16, 2026



Meridian STEM Academy

## Weekly Overview

### Essential Question

How can engineers design a device that converts a renewable energy source into usable electrical energy to solve a real-world problem?

### Learning Targets — I can...

- explain how energy transforms from one form to another within a circuit
- distinguish renewable from nonrenewable sources and evaluate trade-offs
- apply the engineering design process to build and test a working prototype
- construct and troubleshoot a simple series circuit
- communicate design decisions using labeled diagrams and test data

### Standards Alignment

**MS-PS3-3** — design, construct, and test a device that transfers energy

**MS-ETS1-1** — define criteria and constraints of a design problem

**MS-ETS1-2** — evaluate competing design solutions

**MS-ETS1-4** — develop a model to generate data for iterative testing

**CCSS.ELA-WHST.6-8.7** — conduct a short, focused research task

### Cross-Curricular Links

Math (energy-cost ratios) • ELA (engineering logs) • Social Studies (energy policy)

## Materials & Setup Checklist

### Equipment & Tools

- ☐ D-cell batteries & holders (1 per group of 3)
- ☐ Insulated copper wire, pre-stripped, 15 cm lengths
- ☐ Miniature bulbs and sockets, or LEDs with resistors
- ☐ Small DC motors with mini fan blades
- ☐ Multimeters (class set of 6)
- ☐ Alligator-clip test leads

### Consumables & Printables

- ☐ Engineering Design Log packets (1 per student)
- ☐ Renewable Energy Source sorting cards
- ☐ Circuit diagram worksheets
- ☐ Friday showcase rubric handout
- ☐ Recycled-materials bin (cardboard, caps, foil)
- ☐ Safety glasses (class set)

## Daily Lesson Plans

### Monday — Energy Forms & Transfer (Intro)

**Objective:** Students will explain how energy changes form as it moves through a system, using a circuit as a concrete example.

**Standards:** MS-PS3-3

**Warm-Up (5 min):** “Energy Scavenger Hunt” — identify three examples of energy transformation